Math 9 Fall 2022 — MATLAB Homework #1
Corresponding Outcomes: EO1, GOMa2

Instructions: In a group of 1-3 students, create a MATLAB script which does all of the following. Name this file `Face.m`. You must use the exact file name `Face.m` and you must have nothing extra printed to the screen. Put the names of everyone in your group as a comment at the beginning of the file. All members of the group must submit a file for homework, but the file can be the same.

1. Your script should first clear all variables in the workspace using the "clear" command.

2. Find examples of numbers $x_1, x_2, x_3, y_1, y_2, y_3$ such that `plot(`$[x_1, x_2, x_3], [y_1, y_2, y_3]$`)` is different from `plot(`$[x_2, x_1, x_3], [y_2, y_1, y_3]$`)`. Define these numbers and then make both plots. Precede each plot by the command "`figure()`" so that the plots open in new windows.

3. After another "`figure()`" command, plot $y = x^2$ from -5 to 5 using 10 sample points and also using 100 sample points on the same axes.

4. After another "`figure()`" command, plot the exact same thing as in the previous question, but this time zoom in to $-1 \leq x \leq 1$ and $0 \leq y \leq 1$ using the "`axis`" command. In the previous figure, the graphs might look pretty similar, but in the second figure, it's very clear that the plot with 100 sample points is more accurate. At this point, running the script should open four separate figure windows.

5. After another "`figure()`" command, generate the plot below representing a face. It doesn't have to be exactly the same, but it should look approximately the same. In particular, it should have the following aspects:

   (a) The outer circle should be a smooth-looking solid line (so use enough sample points that it looks smooth) of radius 1.

   (b) The two eyes should be made up of 9 or 10 black X's, and they should be the same shape. (Hint. adding or subtracting a number from the x-coordinate shifts the image right or left, respectively.)

   (c) The mouth should be an ellipse shape, longer in the x-direction than in the y-direction (Hint. multiply the x-values to stretch), and it should be made up of many red circles.

   (d) The figure should be titled with the word "Face".

   (e) A legend should be used indicating "head", etc.

   (f) The axis range should be from -1.5 to 1.5 in both the x and y directions.

   (g) Come up with one thing to add to the face to make it unique! (e.g. Maybe the face needs a hat?) Use the following syntax to save the face as an image. Then upload it to the dedicated Ed Discussion thread to compare it with your classmates. (

      - Where you have `figure()`, change it to be `face = figure()`. (You only need to upload one image per group.)
      - At the bottom of your script, write `saveas(face,"face.png","png")`.

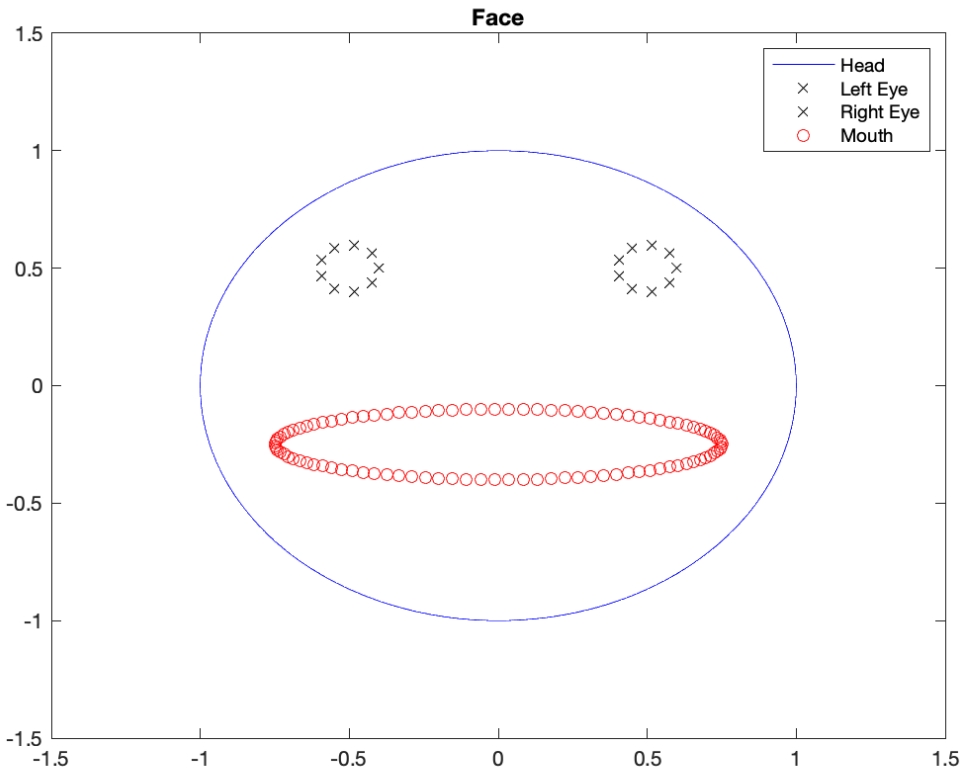

Figure 1: What emotion is this face portraying? Does the face have a name?